

Top view drawing of a rectangular plate. The overall dimensions are 52.70 (width) and 90.80 (height). The inner dimensions are 44.80 ± 0.05 (width) and 82.90 ± 0.05 (height). The plate has four corner features, each consisting of a circular hole with a diameter of 2.0. There are also two locator pillars, each with a diameter of 2.0 and a height of 0.4. A U2 relief pocket is located near the center. The drawing includes dimension lines and arrows pointing to the features.

Dimensions:

- Overall Width: 52.70
- Overall Height: 90.80
- Inner Width: 44.80 ± 0.05
- Inner Height: 82.90 ± 0.05
- Corner Hole Diameter: 2.0
- Locator Pillar Diameter: 2.0
- Locator Pillar Height: 0.4

Features:

- U2 RELIEF POCKET (NOTE 7)
- 2× LOCATOR PILLAR, Ø3.0 × 0.4 TALL (NOTE 4)

4× M2 THREAD, THRU
Ø1.6 TAP DRILL • TAP FROM BACK FACE
Ø3.0 SPOTFACE, BACK (depth per Detail B)

SECTION A-A → edge profile DETAIL B → corner boss

C1	Cavity depth (boss-top plane → cavity floor)	1.80 ±0.05
C2	PCB-rest plane flatness (lip + 4 corner bosses)	FLAT 0.05
C3	4× mounting-hole pattern, pitch 44.80×82.90	±0.05 (linear)
C4	Mounting holes — tapped, thru, from back	M2 / Ø1.6 drill

1. MATERIAL: TITANIUM Gr5 (TC4) = Ti-6Al-4V GRADE 5. 2. FINISH: BEAD-BLAST MATTE (UNIFORM ON THE STEPPED BACK FACE). REAR ART LASER-MARKED IN THE RECESSED BACK FIELD. 3. GENERAL TOLERANCE PER ISO 2768-1 (MEDIUM). C1-C4 TOLERANCED AS LISTED; C1 & C3 = ± 0.05 . DATUMS: A = LEFT EDGE, B = BOTTOM EDGE, C = PCB-REST PLANE. 4. 2x LOCATOR PILLAR: $\varnothing 3.0 \times 0.4$ TALL, STANDING ON THE CAVITY FLOOR AT (13, 35) AND (33, 55) - BOTH WEST OF THE NFC COIL. LEFT AS ISLANDS IN THE SAME CAVITY PASS AS THE BOSSES. LOCATE THE RESIN BRACE (SEPARATE PART) VIA MATCHING $\varnothing 3.2$ RECESSES. FLOOR STAYS A FULL 1.00 (THE (33,55) RECESS IS SLOTTED). MODELED IN THE STEP. 5. EDGE BREAKS - ALL $0.10 \times 45^\circ$, FELT NOT SEEN, MODELED. CALLED OUT ON SEC A-A / DETAIL B: RIM = outer rim (top & bottom); MOUTH = recess mouth (around PCB); LIP = inner (cavity-side) lip edge; FRAME = proud back-frame bottom edges; BOSS = boss (cavity-side) bottom edges (Detail B). BREAK ALL OTHER EXPOSED EDGES $0.10 \times 45^\circ$. FRONT FACE SEATING ON CORNERS. 6. FLOOR IS NOW A TRUE 1.00 (0.95 LOCAL UNDER THE U2 POCKET), AT THE ~ 1.0 TI MIN-WALL GUIDANCE SO IT ALSO CLEARS ALUMINIUM / COPPER / STAINLESS. CAVITY 1.80 $\pm 0.05 \rightarrow$ 1.75 WORST-CASE, MINUS W517 1.70 MAX (DATASHEET CASE W517 HEIGHT) = 0.05 NON-CONTACT; THE BRACE + SOLAR-CELL SANDWICHES CARRY THE BOARD. 7. U2 RELIEF POCKET: 7.8×5.4 CENTERED (28.5, 37) ON THE CAVITY FLOOR, 0.05 DEEP (LOCAL FLOOR 0.95) - U2 (1.75) KEEPS 0.10 AIR. GENERAL FLOOR 1.00. MODELED IN THE STEP. 8. FRONT SUPPORT LIP IS ASYMMETRIC (SECTION A-A): WEST 2.5 / NORTH 2.0 / SOUTH 2.0 FOR PCB RIGIDITY; EAST 1.0 THROUGH THE JP1/TP1 PADS + OVER THE NFC COIL (A GROUNDED LIP WOULD DETUNE IT), WIDENING TO 2.5 AT THE N/S ENDS. THE EXTERIOR BACK BORDER (PLAN) IS INDEPENDENT AND UNIFORM 2.0 ON ALL 4 SIDES. 9. NO INTERNAL RIBS OR SUPPORT POSTS: THE RESIN BRACE (SEPARATE PART) CARRIES CENTER SUPPORT. A PCB LAYOUT CHANGE = A BRACE REPRINT, NOT A SHELL RE-MACHINE. 10. PCB RECESS = 0.60 DEEP (RECEIVES THE 0.60 mm BOARD; SLIP FIT, NOT A PRESS FIT). 3D STEP GOVERNS ALL GEOMETRY; STL NOT FOR CNC. 11. PART = BACK-SHELL ONLY. PCB, RESIN BRACE, AND 4x M2x3 BRASS SLOTTED-CHEESE SCREWS (HEAD $\varnothing 3.8$) SUPPLIED SEPARATELY.		DWG solar-glow-drh-v3_0-backshell-0p6b-brace REV v3.0	
MATERIAL: Ti-6Al-4V (TC4) FINISH: BEAD-BLAST		UNITS mm SCALE AS NOTED BBOX 52.70 x 90.80 x 3.55 3rd-angle	SHEET 1/1



GENERAL TOLERANCES (UNLESS OTHERWISE NOTED, mm)	
LINEAR & HOLE-POSITION DIMS	ISO 2768-m
CAVITY DEPTH C1 (SECTION A-A)	1.80 ±0.05
HOLE-PATTERN PITCH C3 (PLAN)	44.80 / 82.90 ±0.05
PCB-REST FLATNESS C2	FLAT 0.05

SOLAR-GLOW DRH v3.0 — Ti BACK-SHELL (0.6mm-BOARD DUMB BOX)	
DWG solar-glow-drh-v3_0-backshell-0p6b-brace REV v3.0	
MATERIALS (PER DETAIL B) — FINISH BEAD-BLAST	
UNITS mm	SCALE AS NOTED
TERIOR BACK BORDER (PLAN) IS INDEPENDENT AND UNIFORM 2.0 ON ALL 4 SIDES.	
BBOX 52.70 x 90.80 x 3.55 3rd-angle	SHEET 1/1